# Abstract Algebra by Pinter, Chapter 18

Amir Taaki

Chapter 18 on Ideals

## Contents

**A. Examples of Subrings** **2**
Q1 . . . 2
Q2 . . . 3
Q3 . . . 3
Q4 . . . 3
Q5 . . . 3
Q6 . . . 3

**B. Examples of Ideals** **3**
Q1 . . . 3
$\{(n, n) : n \in \mathbb{Z}\}$ . . . 3
$\{(5n, 0) : n \in \mathbb{Z}\}$ . . . 3
$\{(n, m) : n + m$ is even $\}$ . . . 4
$\{(2n, 3m) : n, m \in \mathbb{Z}\}$ . . . 4
Q2 . . . 4
Q3 . . . 4
Q4 . . . 4
Q5 . . . 4
Q6 . . . 4
a . . . 4
b . . . 4
Q7 . . . 5
Q8 . . . 5
Q9 . . . 5

**C. Elementary Properties of Subrings** **5**
Q1 . . . 5
Q2 . . . 5
Q3 . . . 5
Q4 . . . 5
Q5 . . . 5
Q6 . . . 5
Q7 . . . 5
a . . . 5
b . . . 6
Q8 . . . 6
Q9 . . . 6

**D. Elementary Properties of Ideals** **6**
Q1 . . . 6
Q2 . . . 6
Q3 . . . 6
Q4 . . . 7
Q5 . . . 7
Q6 . . . 7

**E. Examples of Homomorphisms** **7**
Q1 . . . 7
Q2 . . . 7
Q3 . . . 7
Q4 . . . 8
Q5 . . . 8
Q6 . . . 8
Q7 . . . 9

**F. Elementary Properties of Homomorphisms** **9**
Q1 . . . 9
Q2 . . . 9
Q3 . . . 9
Q4 . . . 10
Q5 . . . 10
Q6 . . . 10
Q7 . . . 11

**G. Examples of Isomorphisms** **11**
Q1 . . . 11
Addition . . . 11
Multiplication . . . 11
Q2 . . . 11
Addition . . . 11
Multiplication . . . 11
Q3 . . . 12
Q4 . . . 12
Q5 . . . 12

**H. Further Properties of Ideals** **12**
Q1 . . . 12
Q2 . . . 12
Q3 . . . 13
Q4 . . . 13
Q5 . . . 13

**I. Further Properties of Homomorphisms** **13**
Q1 . . . 13
Q2 . . . 14
Q3 . . . 14
Q4 . . . 14
Q5 . . . 14

**J. A Ring of Endomorphisms** **14**
Q1 . . . 14
Q2 . . . 14
Q3 . . . 14
Q4 . . . 14
Q5 . . . 15
Q6 . . . 15

# A. Examples of Subrings

## Q1

$$\{x + \sqrt{3}y : x, y \in \mathbb{Z}\}$$

Closed wrt subtraction

$$(x + \sqrt{3}y)(v + \sqrt{3}w) = xv + \sqrt{3}(yv + xw) + 3yw \in \mathbb{R}$$

Thus it's a subring.

### Q2

As before, it's closed under subtraction and multiplication.

### Q3

$$\{x2^y : x, y \in \mathbb{Z}\}$$

Closed under multiplication because:

$$x_1 2^{y_1} \cdot x_2 2^{y_2} = (x_1 x_2) 2^{y_1 + y_2}$$

Also contains negatives since $x \in \mathbb{Z}$.

To show closure under addition is trivial for positive powers since

$$x2^y + v2^w = x2^{(y-w)}2^w + v2^w = (x2^{(y-w)} + v)2^w$$

Now for the negative case, assume $y > w$, hence $y - w$ is positive and the formulation still holds.

### Q4

The sum and product of continuous functions are continuous.

### Q5

The sum and product on any interval $[0, 1]$ also remains continuous, and hence also includes $\mathcal{C}$

### Q6

Addition and negatives remain in $\mathcal{M}_2(\mathbb{R})$ as does multiplication

$$\begin{pmatrix} 0 & 0 \\ 0 & x \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & xy \end{pmatrix}$$

## B. Examples of Ideals

### Q1

*Identify which of the following are ideals of* $\mathbb{Z} \times \mathbb{Z}$

$\{(n, n) : n \in \mathbb{Z}\}$

$$\begin{aligned} (n, n) + (m, m) &= (m + n, m + n) \in I \\ -(n, n) &= (-n, -n) \in I \\ (n, n) \cdot (a, b) &= (na, nb) \notin I \end{aligned}$$

Not an ideal.

$\{(5n, 0) : n \in \mathbb{Z}\}$

$$\begin{aligned} (5m, 0) + (5n, 0) &= (5(m + n), 0) \in I \\ -(5n, 0) &= (5(-n), 0) \in I \\ (5n, 0) \cdot (a, b) &= (5(na), 0) \in I \end{aligned}$$

Is an ideal.

$\{(n,m) : n+m$ **is even** $\}$

$$\begin{aligned}(n_1,m_1)+(n_2,m_2) &= (n_1+n_2, m_1+m_2) \in I \\ -(n,m) &\in I \\ (n,m)\cdot(a,b) &= (na,mb)\end{aligned}$$

$na$ is even and $mb$ is even, so $na+mb$ is even so $(na,mb) \in I$.

Is an ideal.

$\{(2n,3m) : n,m \in \mathbb{Z}\}$

$$\begin{aligned}(2n_1,3m_1)+(2n_2,3m_2) &= (2(n_1+n_2),3(m_1+m_2)) \in I \\ -(2n,3m) &= (2(-n),3(-m)) \in I \\ (2n,3m)\cdot(a,b) &= (2na,3mb) \in I\end{aligned}$$

Is an ideal

## Q2

*List all the ideals of* $\mathbb{Z}_{12}$

$\mathbb{Z}_{12} = \langle 1 \rangle$ and is cyclic. All subgroups are also cyclic.

- $\mathbb{Z}_{12} =< 1 >=< 5 >=< 7 >=< 11 >$ because $\gcd(m,12) = 1$
- $< 4 >=< 8 >, < 4 >= \{4,8,0\}$
- $< 3 >=< 9 >, < 3 >= \{3,6,9,0\}$
- $< 2 >=< 10 >, < 2 >= \{2,4,6,8,10,0\}$
- $< 6 >= \{6,0\}$
- $< 0 >= \{0\}$

Let $m \in \bar{m} =< m >$, then $< m >= \{mj : j \in \mathbb{Z}_{12}\}$

Let $x \in < m >$ and $y \in \mathbb{Z}_{12}$, since $x \in < m >$, then $x = mj$ for some $j \in \mathbb{Z}_{12}$, thus $xy = mjy$, thus $< m >$ is an ideal of $\mathbb{Z}_{12}$.

Ideals are $< 0 >, < 1 >, < 2 >, < 3 >, < 4 >, < 6 >$

## Q3

See previous exercise

## Q4

$$\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}0 & 0\\ 0 & x\end{pmatrix} = \begin{pmatrix}0 & x\\ 0 & x\end{pmatrix} \notin \mathcal{M}_2(\mathbb{R})$$

## Q5

The product of a continuous and non-continuous function are non-continuous, hence $\mathcal{C}(\mathbb{R})$ is not an ideal of $\mathcal{F}(\mathbb{R})$

## Q6

**a**

Assume he means multiplication here.

$$f(x)\cdot g(x) = 0 \quad \forall x \in \mathbb{Q}$$

Thus $f \cdot g \in I$

**b**

Likewise $f(0)g(0) = 0g(0) = 0$, so $f \cdot g \in I$

### Q7

Ideals of $P_3$ such that $AB = A \cap B \in I$. See also 17D5

$$P_3 = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$$

Any subgroup must contain $\emptyset$.

$A + A = \emptyset$ so $A$ is its own negative.

$\{\emptyset, \{a\}\}$, $\{\emptyset, \{b\}\}$, $\{\emptyset, \{c\}\}$ are all ideals since $\{a\}\{a,c\} = \{a\}$ and $\{a\}\{b,c\} = \emptyset$.

Likewise $\{\emptyset, \{a\}, \{c\}, \{a,c\}\}$, $\{\emptyset, \{a\}, \{b\}, \{a,b\}\}$, $\{\emptyset, \{b\}, \{c\}, \{b,c\}\}$ since $\{a,c\}\{b,c\} = \{c\}$

Lastly we have $P_3$ itself

### Q8

Example of a non-ideal subring is $\{\emptyset, \{a,c\}\}$ which is closed under addition, negatives and multiplication.

### Q9

$$A = \langle (1,1) \rangle = \{(0,0), (1,1), (2,2)\}$$

## C. Elementary Properties of Subrings

### Q1

Let $x \in B$ and since $B$ is a ring then $0 \in B$, thus $0 - x = -x \in B$.

So $B$ is closed wrt negatives and hence addition since $x - (-y) = x + y \in B$

### Q2

As per part 1

### Q3

A ring is a group under addition. Hence order of a subring divides ring by Lagrange.

### Q4

$A$ has no zero divisors, hence neither does $B \implies B$ is an integral domain.

### Q5

$B$ is a subring of field $F$. Let $b \in B, b \neq 0$, then $b^{-1} \in F$ (because $F$ is a field and contains inverses). Every field is an integral domain, hence so is $B$.

### Q6

$F$ is a commutative ring with inverses and unity.

Since $B$ is a subring, it also is commutative.

Since $B$ also contains inverses and is closed wrt multiplication, it must contain $1_F$.

Thus $B$ is a field.

### Q7

**a**

$$B = \langle 2 \rangle = \{0, 2, 4, \ldots, 16\}$$

**b**

$$B = \langle 9 \rangle = \{0, 9\}$$

### Q8

$$f(e) = e$$
$$f(x_1 + x_2) = f(x_1) + f(x_2)$$
$$f(x_1 x_2) = f(x_1) f(x_2)$$

But $\forall x \in B \quad f(x) = x$

$$x_1, x_2 \in B$$
$$x_1 + x_2 = f(x_1) + f(x_2) = f(x_1 + x_2)$$

Likewise for multiplication.

Since $A$ is a ring $\forall - x \in A$ st $x + (-x) = e$ but $x \in B$

$$f(x) + f(-x) = f(e) = f(x + (-x)) = x + (-x)$$

Hence $-x \in B$ also.

### Q9

$$ax = xa \qquad bx = xb$$
$$(a + b)x = x(a + b)$$

So $a + b$ also is in the center.

$$(ab)x = axb = x(ab)$$

Finally $0x = 0 = x0$

$$-a \in A$$
$$-ax = -(ax) = -(xa) = -xa$$

By associativity.

## D. Elementary Properties of Ideals

### Q1

*Explain why $J$ is an ideal of $A$ iff $J$ is closed with respect to subtraction and $J$ absorbs products in $A$.*

$$0 - x = -x \in J$$
$$x - (-y) = x + y \in J$$

So $J$ is closed wrt negatives and addition from the statement about subtraction.

### Q2

*If $A$ is a ring with unity, prove that $J$ is an ideal of $A$ iff $J$ is closed with respect to addition and $J$ absorbs products in $A$.*

Note that $A$ is a ring with unity, and by definition must include $-1$.

Then note that since $J$ absorbs products, that $(-1) \cdot a = -a \in J$.

### Q3

*Prove that the intersection of any two ideals of $A$ is an ideal of $A$.*

1. Since $x, y \in I_j$, and $I_j$ is an ideal, $x - y \in I_j, \forall j \in J$. Therefore $x - y \in \bigcap_{j \in J} I_j = I$.
2. Since $x \in I_j, rx \in I_j, \forall j \in J$. Therefore $rx \in I$.

### Q4

*Prove that $J$ is an ideal of $A$ and $1 \in J$, then $J = A$.*

Since ideals absorb products, then if $1 \in J$, then since $a \cdot 1 = a \in J$, then $J = A$.

### Q5

*Prove that if $J$ is an ideal of $A$ and $J$ contains an invertible element $a$ of $A$, then $J = A$.*

$$a \cdot a^{-1} = 1 \in J$$

By previous exercise $J = A$.

### Q6

*Explain why a field $F$ can have no nontrivial ideals.*

Every nonzero element of a field is invertible. Hence the only ideals are $\{0\}$ or $F$ itself.

## E. Examples of Homomorphisms

### Q1

Let $f, g \in \mathcal{F}(\mathbb{R})$

$$\phi(f+g) = (f+g)(0) = f(0) + g(0) = \phi(f) + \phi(g)$$
$$\phi(f \cdot g) = (f \cdot g)(0) = f(0)g(0) = \phi(f)\phi(g)$$

$$K = \{f \in \mathcal{F}(\mathbb{R}) : f(0) = 0\}$$

Range is $[-\infty, \infty]$.

### Q2

$$h : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$
$$h(x, y) = x$$
$$h(x_1 + x_2, y_1 + y_2) = x_1 + x_2 = h(x_1, y_1) + h(x_2, y_2)$$
$$h(x_1 x_2, y_1 y_2) = x_1 x_2 = h(x_1, y_1)h(x_2 y_2)$$

$$\begin{aligned} K &= \{x, y \in \mathbb{R} \times \mathbb{R} : h(x, y) = 0\} \\ &= \{(0, y) : y \in \mathbb{R}\} \end{aligned}$$

Range is $[-\infty, \infty]$

### Q3

$$h : \mathbb{R} \to \mathcal{M}_2(\mathbb{R})$$
$$h(x) = \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}$$
$$h(x+y) = \begin{pmatrix} x+y & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} y & 0 \\ 0 & 0 \end{pmatrix} = h(x) + h(y)$$
$$h(xy) = \begin{pmatrix} xy & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} y & 0 \\ 0 & 0 \end{pmatrix} = h(x)h(y)$$
$$K = \{0\}$$

Range is

$$\begin{pmatrix} \pm\infty & 0 \\ 0 & 0 \end{pmatrix}$$

## Q4

$$h : \mathbb{R} \times \mathbb{R} \to \mathcal{M}_2(\mathbb{R})$$
$$h(x, y) = \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}$$

$$\begin{aligned} h(x_1 + x_2, y_1 + y_2) &= \begin{pmatrix} x_1 + x_2 & 0 \\ 0 & y_2 + y_2 \end{pmatrix} \\ &= \begin{pmatrix} x_1 & 0 \\ 0 & y_1 \end{pmatrix} + \begin{pmatrix} x_2 & 0 \\ 0 & y_2 \end{pmatrix} \\ &= h(x_1, y_1) + h(x_2, y_2) \end{aligned}$$

$$K = \{(0, 0)\}$$

Range is

$$\begin{pmatrix} \pm\infty & 0 \\ 0 & \pm\infty \end{pmatrix}$$

## Q5

$$f : \mathbb{R} \times \mathbb{R} \to \mathcal{M}_2(\mathbb{R})$$
$$f(x, y) = \begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix}$$

$$\begin{aligned} f(x_1 + x_2, y_1 + y_2) &= \begin{pmatrix} x_1 + x_2 & 0 \\ y_2 + y_2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} x_1 & 0 \\ y_1 & 0 \end{pmatrix} + \begin{pmatrix} x_2 & 0 \\ y_2 & 0 \end{pmatrix} \\ &= f(x_1, y_1) + f(x_2, y_2) \end{aligned}$$

$$f((x_1, y_1) \otimes (x_2, y_2)) = f(x_1 x_2, y_1 x_2) = \begin{pmatrix} x_1 x_2 & 0 \\ y_1 x_2 & 0 \end{pmatrix}$$
$$f(x_1, y_1) f(x_2 y_2) = \begin{pmatrix} x_1 & 0 \\ y_1 & 0 \end{pmatrix} \begin{pmatrix} x_2 & 0 \\ y_2 & 0 \end{pmatrix} = \begin{pmatrix} x_1 x_2 & 0 \\ y_1 x_2 & 0 \end{pmatrix}$$
$$K = \{(0, 0)\}$$

## Q6

$$h : P_C \to P_C$$
$$h(A) = A \cap D$$
$$D \subseteq C$$

$$\begin{aligned} h(A + B) &= h((A - B) \cup (B - A)) \\ &= [(A - B) \cup (B - A)] \cap D \\ &= [(A - B) \cap D] \cup [(B - A) \cap D] \\ &= [A \cap D - B \cap D] \cup [B \cap D - A \cap D] \\ &= h(A) + h(B) \\ h(AB) &= h(A \cap B) = A \cap B \cap D \\ &= (A \cap D) \cap (B \cap D) \\ &= h(A) h(B) \end{aligned}$$

$$K = \{A \in P_C : A \cap D = \emptyset\}$$

Range is every subset of $D$.

### Q7

Rules for ring homomorphisms:

$$f(a+b) = f(a) + f(b) \qquad f(ab) = f(a)f(b)$$

$$f(0) = 0 \qquad f(1_A) = 1_B$$

$$f(n) = f(1 + \cdots + 1) = f(1) + \cdots + f(1) = nf(1)$$

$$mf(1) = 0 \qquad f(1)^2 = f(1)$$

Homomorphisms for $\phi_i : \mathbb{Z}_2 \to \mathbb{Z}_4$

$$\phi_e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

The other mappings do not work:

- $1 \to 1$ then $2f(1) \neq 0$
- $1 \to 2$ then $f(1)^2 \neq f(1)$
- $1 \to 3$ then $f(1)^2 \neq f(1)$

Homomorphisms for $\phi_i : \mathbb{Z}_2 \to \mathbb{Z}_4$

$$\phi_e = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\phi_a = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 4 & 2 \end{pmatrix}$$

The other mappings do not work:

- $1 \to 1$ then $f(1)^2 \neq f(1)$
- $1 \to 2$ then $f(1)^2 \neq f(1)$
- $1 \to 3$ then $3f(1) \neq 0$
- $1 \to 5$ then $3f(1) \neq 0$

## F. Elementary Properties of Homomorphisms

### Q1

*Prove $f(A) = \{f(x) : x \in A\}$ is a subring of $B$.*

Since $f$ is a homomorphism, ring operations are obeyed in the homomorphism. For negatives we note that $f(0_A) = 0_B = 1_B + (-1_B)$ and every negative is expressible as $(-1_B) \cdot a$ where $a \in B$.

### Q2

*Prove the kernel of $f$ is an ideal of $A$.*

$$K = \{x \in A : f(x) = 0_B\}$$

From $f$ being a homomorphism, we conclude $K$ is a subring of $A$.

To show it's an ideal, for any $a \in A$ and $x \in K$, then $f(ax) = 0 = f(x)$. So $K$ absorbs the product $ax$.

Thus the kernel of a homomorphism is an ideal of the input ring.

### Q3

*Prove $f(0) = 0$, and for every $a \in A, f(-a) = -f(a)$.*

$$f(0) = f(0+0) = f(0) + f(0) \implies f(0) = 0$$

$$f(a + (-a)) = f(a) + f(-a) = f(0) = 0 = f(a) - f(a)$$

$$\implies f(-a) = -f(a)$$

## Q4

*Prove $f$ is injective iff its kernel is equal to $\{0\}$.*

$$f(x) = f(y) \iff f(y-x) = 0 \iff y - x \in K$$

Let $x \in K$

$\implies f(x) = 0$

$\implies f(x) = f(0) \qquad$ [since $f(0) = 0$]

$\implies x = 0 \qquad$ [since $f$ is injective]

It follow $K = \{0\}$

Thus $f$ is injective $\implies K = \{0\}$

Now suppose $K = \{0\}$. Then

$f(x) = f(y)$

$\implies f(x) - f(y) = 0$

$\implies f(x - y) = 0$

$\implies x - y \in K$

$\implies x - y = 0 \qquad$ [since $K = \{0\}$]

$\implies x = y$

Hence $f$ is injective.

Thus $K = \{0\} \implies f$ is injective.

Hence $f$ is injective $\iff K = \{0\}$

## Q5

*If $B$ is an integral domain, then either $f(1) = 1$ or $f(1) = 0$. If $f(1) = 0$ then $f(x) = 0$ for every $x \in A$. If $f(1) = 1$, the image of every invertible element of $A$ is an invertible element of $B$.*

Integral domain has the cancellation property such that $ab = ac \implies b = c$.

$$f(1) = f(1 \cdot 1) = f(1)f(1)$$
$$f(1) = f(1)f(1)$$

$f(1) = 0$ or 1$

If $f(1) = 0$ then $\forall a \in A$, $f(a) = f(1 \cdot a) = f(1)f(a) = 0$

If $f(1) = 1$ and $\exists x, y \in A$ such that $xy = 1$

$$f(xy) = f(x)f(y) \text{ where } f(y) = (f(x))^{-1}$$

## Q6

*Any homomorphic image of a commutative ring is a commutative ring. Any homomorphic image of a field is a field.*

Let $a, b \in A$, then $f(a)f(b) = f(b)f(a)$ because $f(ab) = f(ba)$.

If $A$ is a field, then $\forall x \in A$, $\exists x^{-1} \in A$. So by the last exercise, $f(x^{-1}) = (f(x))^{-1}$ and so the inverse of $f(x)$ is a member of $B$.

$$(f(x))^{-1} \in B$$

### Q7

*If the domain $A$ of the homomorphism $f$ is a field, and if the range of $f$ has more than one element, then $f$ is injective.*

Since $A$ is a field, the kernel of $A$ is either $\{0\}$ or $A$ itself.

But the range of $f$ is more than one element, so the kernel of $A$ cannot be $A$ and must be $\{0\}$.

Since the kernel of $f$ is $\{0\}$, then $f$ is injective.

## G. Examples of Isomorphisms

### Q1

$$a \oplus b = a + b + 1$$
$$a \otimes b = ab + a + b$$

#### Addition

$$\begin{aligned} f(a+b) &= a + b - 1 \\ f(a) \oplus f(b) &= (a-1) + (b-1) - 1 \\ &= a + b - 1 \end{aligned}$$

#### Multiplication

$$\begin{aligned} f(ab) &= ab - 1 \\ f(a) \otimes f(b) &= (a-1)(b-1) + (a-1) + (b-1) \\ &= ab - b - a + 1 + a + b - 1 - 1 \\ &= ab - 1 \end{aligned}$$

### Q2

$$\mathcal{J} = \{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a, b \in \mathbb{R} \}$$
$$f : \mathbb{C} \to \mathcal{J}$$
$$f(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$
$$a + bi = c + di \implies f(a + bi) = f(c + di)$$

#### Addition

$$f((a + bi) + (c + di)) = \begin{pmatrix} a + c & b + d \\ -(b + d) & a + c \end{pmatrix}$$
$$f(a + bi) + f(c + di) = \begin{pmatrix} a + c & b + d \\ -(b + d) & a + c \end{pmatrix}$$

#### Multiplication

$$\begin{aligned} f((a + bi)(c + di)) &= f((ac - bd) + (ad + bc)i) \\ &= \begin{pmatrix} a + c & b + d \\ -(b + d) & a + c \end{pmatrix} \\ f(a + bi) f(c + di) &= \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} c & d \\ -d & c \end{pmatrix} \\ &= \begin{pmatrix} a + c & b + d \\ -(b + d) & a + c \end{pmatrix} \end{aligned}$$

### Q3

$$A = \{(x, x) : x \in \mathbb{Z}\}$$

$\forall x, y \in \mathbb{Z}, (x, x) \in A, (y, y) \in A, (x + y, x + y) \in A$ and $(xy, xy) \in A$

Thus $A$ is a subring of $\mathbb{Z} \times \mathbb{Z}$

The homomorphism $f : \mathbb{Z} \to A$ by $f(x) = (x, x)$ is isomorphic because it is one to one

$$f(x) = f(y) \implies x = y$$

and onto

$$\forall (x, x) \in A, \exists x \in \mathbb{Z} \text{ such that } f(x) = (x, x)$$

Thus

$$\{(x, x) : x \in \mathbb{Z}\} \cong \mathbb{Z}$$

### Q4

Addition and negatives trivially remain inside the set.

$$\begin{pmatrix} 0 & 0 \\ 0 & x \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & xy \end{pmatrix}$$

Hence the set is a subring.

$$A = \{\begin{pmatrix} 0 & 0 \\ 0 & x \end{pmatrix} : x \in \mathbb{R}\}$$

Define $f : \mathbb{R} \to A$ by $f(x) = \begin{pmatrix} 0 & 0 \\ 0 & x \end{pmatrix}$, then $f$ is an homomorphism from $\mathbb{R}$ to $A$.

Hence $A \cong \mathbb{R}$

### Q5

$$f : k\mathbb{Z} \to l\mathbb{Z}$$

$$f(k) = ln \text{ for some } n \neq 0$$

$$\begin{aligned} f(k^2) &= l^2n^2 \\ &= f(k \cdot k) = f(k + \cdots + k) = kf(k) \\ &= kln \\ k &= ln \end{aligned}$$

But $k \neq l$, so $ln$ does not generate $l\mathbb{Z}$ and $f$ is not an isomorphism.

## H. Further Properties of Ideals

### Q1

*If $J \cap K = \{0\}$, then $jk = 0$ for every $j \in J$ and $k \in K$.*

$J$ and $K$ are ideals, so for every $j \in J$ and $k \in K$, then $jk \in J$ and $jk \in K$, so $jk \in J \cap K$.

### Q2

*For any $a \in A$, $I_a = \{ax + j + k : x \in A, j \in J, k \in K\}$ is an ideal of $A$.*

$\forall i \in I_a$, and $b \in A$, then $bi = b(ax + j + k) = a(bx) + bj + bk \in I_a$ since $J$ and $K$ are ideals and $bx \in A$.

## Q3

*The radical of $J$ is the set* $\operatorname{rad} J = \{a \in A : a^n \in J \text{ for some } n \in \mathbb{Z}\}$. *For any ideal $J$,* $\operatorname{rad} J$ *is an ideal of $A$.*

$$a^n \in J \qquad b^m \in J$$
$$(a+b)^{n+m} \in J \qquad \text{[see 17m3]}$$

$x \in A$ and $a \in \operatorname{rad} J$ then $(xa)^n = x^n a^n \in J$, so $xa \in \operatorname{rad} J$.

$a, b \in \operatorname{rad} J$, then $(a+b)^{m+n} \in J$ and so $a + b \in \operatorname{rad} J$.

## Q4

*For any $a \in A$, $\{x \in A : ax = 0\}$ is an ideal (called the annihilator of $a$).*

*Furthermore, $\{x \in A : ax = 0$ for every $a \in A\}$ is an ideal (called the annihilating ideal of $A$). If $A$ is a ring with unity, its annihilating ideal is equal to $\{0\}$.*

Let $b \in A$, then $bx \in Ann(a)$ because $ax = 0$ so $b(ax) = bxa = 0$.

Let $x, y \in Ann(a)$ then $a(x+y) = 0$ so $x + y \in Ann(a)$.

$$I = \{x \in A : ax = 0 \text{ for every } a \in A\}$$

If $A$ is a ring with unity then $a = 1 \implies x = 0$ so $I = \{0\}$.

## Q5

*Show that $\{0\}$ and $A$ are ideals of $A$. (They are trivial ideals; every other ideal of $A$ is a proper ideal.) A proper $J$ of $A$ is called maximal if it is not strictly contained in any strictly larger proper ideal: that is if $J \subseteq K$, where $K$ is an ideal containing some element not in $J$, then necessarily $K = A$.*

*Show the following is an example of a maximal ideal: in $\mathcal{F}(\mathbb{R})$, the ideal $J = \{f : f(0) = 0\}$.*

$$g \in K \qquad g(0) \neq 0 \qquad g \notin J$$

$$h(x) = g(x) - g(0) \in J$$
$$h(x) - g(x) \in K$$

Continuous function with a nonzero value is invertible.

$h(x) - g(x) = -g(0) \in K$ but $g(0) \neq 0$ so $-1/g(0) \in A$.

But since $K$ is an ideal, that is

$$g(0) \cdot 1/g(0) \in K$$

but this equals 1, and $1 \in K$ so $K = A$ and is maximal.

# I. Further Properties of Homomorphisms

## Q1

*If $f : A \to B$ is a homomorphism from $A$ onto $B$ with kernel $K$, and $J$ is an ideal of $A$ such that $K \subseteq J$, then $f(J)$ is an ideal of B$.*

$f$ is onto $\exists x : f(x) = y$ so it's an ideal. Closed under addition and negatives and absorbs products.

See also here

### Q2

*If $f : A \to B$ is a homomorphism from $A$ onto $B$, and $B$ is a field, then the kernel of $f$ is a maximal ideal.*

The kernel $K$ is a subset of the ideal for $A$. As shown above $f(J)$ is an ideal of $B$, which by D6 can only be $\{0\}$ or $B$ itself. Since the homomorphism is onto, then $f(A)$ maps to $B$, but $A$ is a trivial ideal of $A$. Thus $K$, the kernel of $f$ is the proper ideal for $A$ which maps to $\{0\}$ in $B$.

### Q3

*There are no nontrivial homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}$.*

$$f(1) = f(1 \cdot 1) = f(1) \cdot f(1)$$
$$f(1) = 1 \text{ or } f(1) = 0$$
$$f(n) = f(1 + \cdots + 1) = f(1) + \cdots + f(1) = nf(1)$$

So $f(n) = n$ or $f(n) = 0$

See also here and here

### Q4

*If $n$ is a multiple of $m$, then $\mathbb{Z}_m$ is a homomorphic image of $\mathbb{Z}_n$.*

$f : \mathbb{Z}_n \to \mathbb{Z}_m\$$ by $f(a) = a (\mathrm{mod}\, m)$ obeys the homomorphic properties.

See also here

### Q5

*If $n$ is odd, there is an injective homomorphism from $\mathbb{Z}_2$ into $\mathbb{Z}_{2n}$.*

$$f(x) = nx$$

Above homomorphism is injective since $f(0) = 0$ and $f(1) = n$.

## J. A Ring of Endomorphisms

### Q1

$$\pi_a(x) = ax$$
$$\pi_a(x + y) = a(x + y) = ax + ay = \pi_a(x) + \pi_a(y)$$

### Q2

$$\pi_a(x) = \pi_a(y) \implies x = y$$

$a$ is not a divisor of zero $\implies \forall x \in A, ax \neq 0$, thus ring $A$ has cancellation property

$$\pi_a(x) = \pi_a(y) = ax = ay \implies x = y$$

### Q3

If $a$ is invertible then $\forall y \in A$, $y = a(a^{-1}y)$ so $x = a^{-1}y$, $f(x) = y$, thus $\pi_a$ is surjective.

### Q4

$$\mathcal{A} = \{\pi_a : a \in A\}$$
$$[\pi_a + \pi_b](x) = \pi_a(x) + \pi_b(x)$$
$$\pi_a\pi_b = \pi_a \cdot \pi_b$$

1. Addition is abelian
2. Multiplication is associative: $(\pi_a \cdot \pi_b \cdot \pi_c)(x) = (abc)x = a(bcx) = \pi_a((\pi_b \cdot \pi_c)(x))$
3. Distributive over addition

## Q5

$$\phi : A \to \mathcal{A} \text{ given by } \phi(a) = \pi_a$$

As shown above this is homomorphic.

## Q6

$$\phi(a) = \phi(b) \implies \pi_a = \pi_b$$
$$\pi_a(1) = \pi_b(1) \implies a = b$$

$\forall \pi_a \in \mathcal{A}, \exists a \in A : \pi_a = \phi(a)$ by definition.

If $a$ has no divisors of zero, then to show injective property, note that

$$ax = bx \implies a = b$$

$$\pi_a = pi_b \implies \pi_a(x) = ax = \pi_b(x) = bx \implies a = b$$

From the cancellation property since it has no divisors of zero.